Week 2 Report: DINO from First Principles

A compact self-supervised Vision Transformer implementation on CIFAR-10

1. What this was about

This notebook implements a small version of DINO (Self-Distillation with No Labels). A student network learns to match predictions from a teacher network across different augmented views of the same image. The student learns through backpropagation; the teacher follows the student using an exponential moving average (EMA) of its weights. The notebook also compares this setup against a no-EMA ablation.

2. What was built

The pipeline includes CIFAR-10 subset selection, multi-crop augmentation, a small Vision Transformer (ViT), a DINO projection head and loss, teacher centering and sharpening, EMA updates, training diagnostics, weighted k-NN evaluation, PCA/t-SNE embedding plots, attention maps, and an ablation that removes EMA.

Component	Notebook configuration
Dataset	5,000 training images; 2,000 test images for evaluation
Backbone	ViT; 4×4 patches, dimension 192, 6 blocks, 3 attention heads
Parameters	Backbone: 2.69M; projection head: 0.69M
Views	2 global crops (32×32) and 4 local crops (16×16)
Training	20 epochs; batch size 64; AdamW; learning rate 2×10■■; weight decay 0.04
DINO settings	Output dimension 2,048; student temperature 0.1; teacher temperature 0.04
Evaluation	Weighted k-NN, PCA/t-SNE and CLS attention maps

Labels are excluded from the training objective. They are used only after training for k-NN evaluation and to colour the embedding plots.

3. How the method works

Multi-crop. Every image produces two large global views and four smaller local views. The teacher sees the global crops; the student sees all six. Matching predictions across views encourages representations that are less dependent on exact crop or colour.

Student and teacher. Both networks start with identical weights. Only the student receives gradients. The teacher is updated using teacher ← m·teacher + (1–m)·student, where momentum m increases from about 0.99 towards 0.9999.

Projection head and loss. The ViT CLS representation passes through an MLP head that produces 2,048 logits. The teacher probabilities are centred and sharpened using temperature 0.04; the student uses temperature 0.1. Cross-entropy compares the teacher and student distributions across different crop pairs.

Collapse prevention. Collapse occurs when outputs become uninformative, for example nearly uniform across the output dimensions. Centering subtracts a running average of teacher outputs, while sharpening encourages more decisive predictions. Entropy and student–teacher KL are tracked to diagnose stability.

4. Implementation checks

The notebook verified the expected output shapes for a batch of 64: 384 student outputs (64 × 6 crops) and 128 teacher outputs (64 × 2 global crops). It also checked that student and teacher initially agree, gradients reach the student but not the teacher, and the EMA update matches its formula. Both sanity checks passed.

5. Main training results

With EMA enabled, loss increased from 7.014 in epoch 1 to 7.287 in epoch 3, then declined to 6.659 by epoch 20. Teacher entropy ended at 5.88, below the maximum ln(2048) ≈ 7.62. The early rise in loss therefore needs to be interpreted alongside entropy and KL rather than on its own.

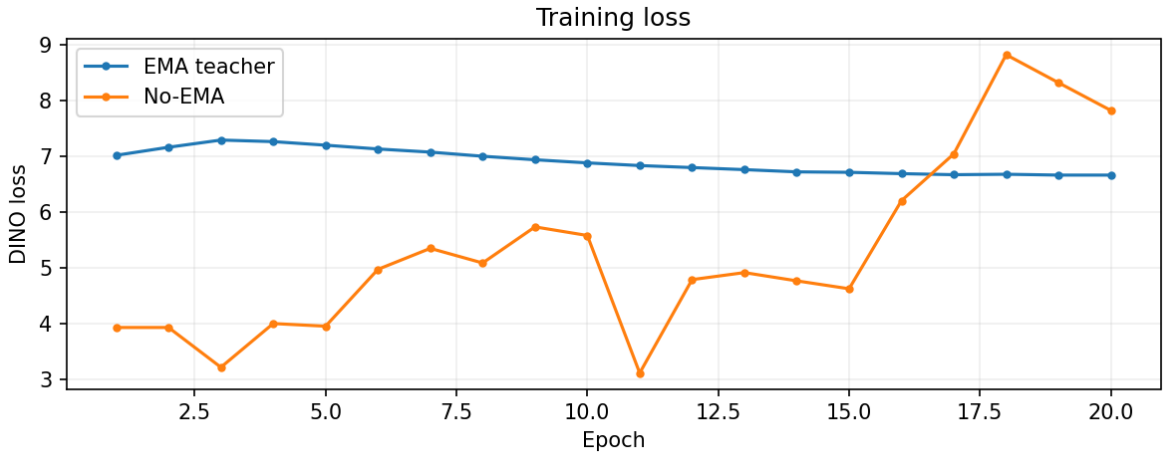


Figure 1. The no-EMA loss becomes unstable and rises sharply near the end of training.

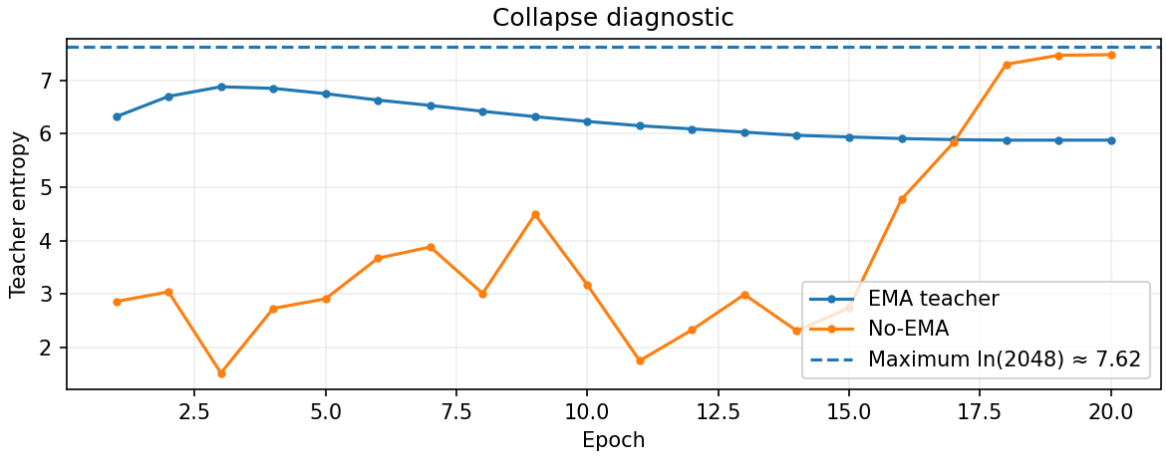


Figure 2. No-EMA teacher entropy approaches the uniform-output maximum.

6. Feature quality and visualisations

Evaluation	Untrained ViT	EMA teacher	Student
Weighted k-NN accuracy	25.7%	28.9%	30.6%

The trained student improved by 4.9 percentage points over the untrained baseline, and the teacher improved by 3.2 points. This is a modest gain rather than strong class separation; the evaluation uses only 2,000 test images and one seed.

PCA and t-SNE. These methods reduce the 192-dimensional embeddings to two dimensions for plotting. The notebook plots show substantial overlap, with only rough separation between some vehicle and animal classes. These plots are qualitative; k-NN gives a more direct numerical measure.

Attention maps. CLS-token attention from the final Transformer block is visualised for each of the three heads. Some maps focus on object regions, but attention does not consistently trace object boundaries and can highlight background regions. The evidence is suggestive, not conclusive.

7. Ablation: removing EMA

In the ablation, the teacher is copied directly from the student at every step ($m = 0$). At epoch 20, the no-EMA run had loss 7.818, entropy 7.48 (maximum ≈ 7.62), and k-NN accuracy 21.3% for both networks. The EMA run finished with loss 6.659, entropy 5.88, and k-NN scores of 28.9% for the teacher and 30.6% for the student.

Metric	EMA (epoch 20)	No EMA (epoch 20)
Loss	6.659	7.818
Teacher entropy	5.88	7.48 / 7.62 max
k-NN accuracy	28.9% teacher; 30.6% student	21.3% both
Weight distance	1.07	0.00

Without EMA, the training signals fluctuate and entropy approaches its maximum, indicating near-uniform teacher outputs. The k-NN score also falls below the untrained baseline. This comparison supports the importance of the EMA teacher for stable training in this implementation.

8. Limitations and takeaway

This is a small educational implementation, not a full reproduction of the DINO paper: it uses 5,000 training images, a 2.69M-parameter backbone, a 2,048-dimensional output, 20 epochs, and a single seed. Performance is sensitive to hyperparameters and subset choice. The key takeaway is that matching a teacher is not enough by itself: EMA, centering, sharpening, and multi-view augmentation must work together to avoid collapse and learn useful features.

Reference: Caron, M. et al. (2021). “Emerging Properties in Self-Supervised Vision Transformers.” Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). This report describes the supplied Week2.ipynb implementation; numerical results are taken from its saved outputs.